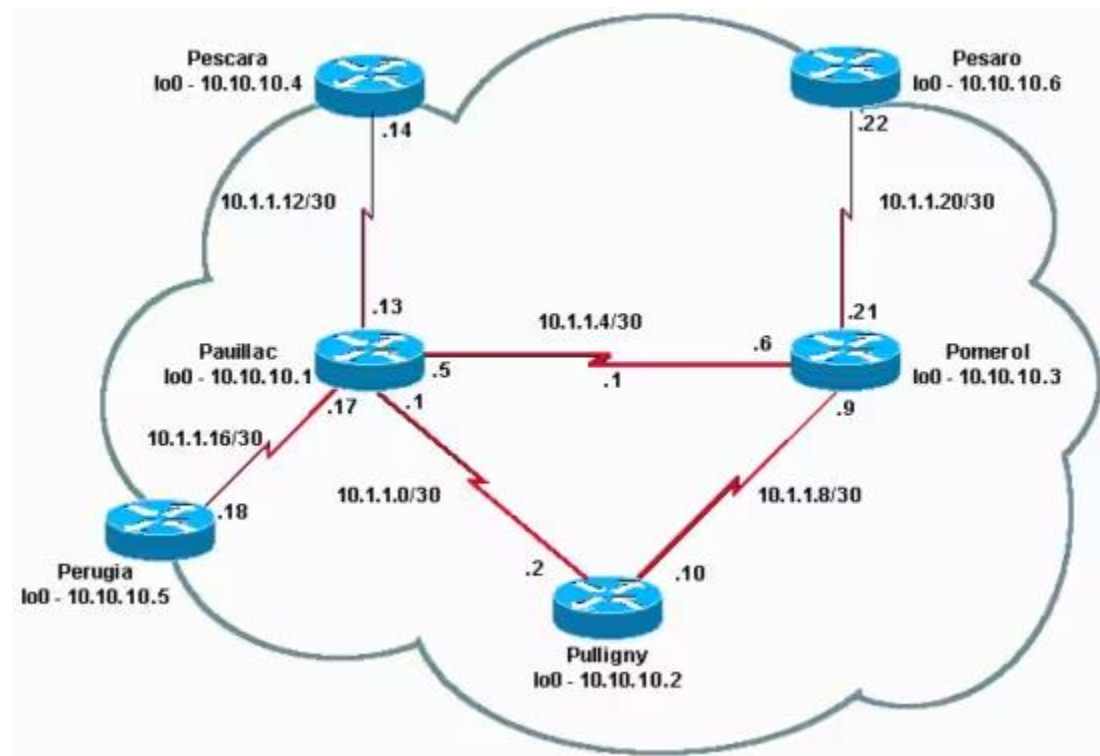


Welcome to our MPLS lab!

In this lab, you will learn about Multiprotocol Label Switching (MPLS), one of the most widely used technologies in modern networking. The lab is divided into six parts, each with its own set of tasks.

Practice Lab tasks



PART 1: Cisco Express Forwarding (CEF)

Task 1.1: Verify CEF Configuration

- Access a Cisco router via the console
- Verify that CEF is enabled on the router
- Verify that the router is running a supported Cisco IOS version

Task 1.2: Configure CEF

- Enable CEF on the router
- Configure CEF load balancing to ensure efficient use of network resources
- Verify CEF operation

PART 2: The MPLS Label

Task 2.1: Understand the MPLS Label

- Learn about the structure of the MPLS label
- Understand how MPLS labels are assigned and used to switch packets

Task 2.2: Assign MPLS Labels

- Assign MPLS labels to IP addresses
- Verify MPLS label assignment

PART 3: The MPLS Components

Task 3.1: Configure Label Distribution Protocol (LDP)

- Configure LDP on the router
- Verify LDP operation

Task 3.2: Understand the LER and LSR

- Learn about the functions of the Label Edge Router (LER) and the Label Switch Router (LSR)

PART 4: The MPLS Process

Task 4.1: Understand the MPLS Process

- Learn about the process of MPLS label distribution and packet switching
- Understand the role of the LDP in MPLS

Task 4.2: Verify MPLS Operation

- Verify MPLS label distribution and packet switching
- Use traceroute to verify the MPLS path

PART 5: Configuring Basic MPLS

Task 5.1: Configure MPLS

- Enable MPLS on the router
- Assign MPLS labels to IP addresses
- Verify MPLS operation

Task 5.2: Configure Traffic Engineering (TE)

- Configure MPLS TE to optimize network traffic
- Verify MPLS TE operation

PART 6: MPLS VPNs

Task 6.1: Configure MPLS VPNs

- Configure MPLS VPNs to provide secure, isolated networks for different customers
- Verify MPLS VPN operation

Task 6.2: Configure Quality of Service (QoS) for MPLS VPNs

- Configure QoS to ensure that voice and video traffic are given priority
- Verify QoS operation

In conclusion, this MPLS lab provides hands-on experience with MPLS technology, including CEF, MPLS labels, MPLS components, MPLS process, configuring basic MPLS, and MPLS VPNs. By completing these tasks, you will have a solid understanding of MPLS and the ability to configure and troubleshoot MPLS networks.

Case Study

Welcome to the Case Study for our MPLS Lab. In this case study, we will use all the concepts learned in the MPLS lab to design a complex network system.

The Scenario:

XYZ Corp is a multinational company that has offices in different parts of the world. To ensure efficient communication and secure data transfer between these offices, they want to implement a network system using MPLS technology.

**** Consider at least 3 Countries**

**** Consider at least 4 departments (Accounting, Operations, HR & Administration, and Sales)**

The network system should meet the following requirements:

- The network should be scalable to accommodate future expansion.
- All offices should have secure connectivity with each other.
- All offices should have access to a centralized data center for shared resources.
- The network should prioritize voice and video traffic to ensure optimal performance.

Company Solution:

We propose a network system that consists of a core network, distribution networks, and access networks. The core network will be the backbone of the entire system, and it will use MPLS to provide secure connectivity between the offices. The distribution networks will connect the access networks to the core network, while the access networks will connect individual devices to the distribution networks.

Meeting minutes from 6 different team members in the network department.

Part 1: Cisco Express Forwarding (CEF)

We will enable CEF on all routers to ensure fast packet switching and efficient use of network resources.

Part 2: The MPLS Label

We will assign MPLS labels to IP addresses to facilitate MPLS packet switching.

Part 3: The MPLS Components

We will configure Label Distribution Protocol (LDP) on all routers to enable label distribution and switching. We will also designate some routers as Label Edge Routers (LERs) and others as Label Switch Routers (LSRs) to ensure efficient packet switching.

Part 4: The MPLS Process

We will verify MPLS label distribution and packet switching and use traceroute to confirm the MPLS path.

Part 5: Configuring Basic MPLS

We will configure MPLS on all routers and assign MPLS labels to IP addresses. We will also configure Traffic Engineering (TE) to optimize network traffic.

Part 6: MPLS VPNs

We will configure MPLS VPNs to provide secure, isolated networks for different offices. We will also configure Quality of Service (QoS) to prioritize voice and video traffic.

The final network system will look something like this:

The core network will consist of multiple routers connected to each other through MPLS.

Each office will have a distribution network connected to the core network through MPLS. The distribution network will consist of multiple routers.

Each device in the office will be connected to the access network, which will be connected to the distribution network through MPLS.

In conclusion, our proposed MPLS-based network system will meet all the requirements of XYZ Corp. It will provide secure connectivity, efficient communication, and optimized performance. The system will also be scalable to accommodate future expansion.